

Computer Network

I. Draw the Network Topology :-

↳ LAN Topology used: Star Topology

↳ Components in Star Topology :-

- (I) Router
- (II) Switch
- (III) Access point (Wifi Router)
- (IV) PCs
- (V) Laptops
- (VI) Mobile Devices

↳ Explanation :

- (I) All devices are connected through a central switch/router.
- (II) If one cable fails, other devices continue working.
- (III) Commonly used in colleges, offices and homes.

Q2. Identify Network Devices ?

Device	Function
↳ Router	↳ Connects LAN to internet
↳ Switch	↳ Connects Multiple devices in LAN.
↳ Access point	↳ Provides wireless wifi connect.
↳ PC	↳ Used by users to access net work
↳ Laptop	↳ Portable device connected through wifi/LAN.
↳ Ethernet Cable	↳ Used for wired communication Cables.

Q3. Analyse Connection type :-

(A) Wired Connection (Ethernet)

- ↳ Uses ethernet cable
- ↳ Faster and Stable connection
- ↳ Mostly used in :

Computer labs.

Offices.

Servers

(B) Wireless Connection (Wifi)

- ↳ Uses radio signals
- ↳ Portable and Flexible
- ↳ Mostly used in :

↳ Mobile phones

↳ laptops

↳ tablets

Ethernet adapter Ethernet 4:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::c629:1ac2:2069:65ca%5
IPv4 Address. : 192.168.56.1
Subnet Mask : 255.255.255.0
Default Gateway :

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Ethernet adapter Bluetooth Network Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

```
KUMAR>getmac
```

```
Transport Name
```

```
=====
```

```
Device\Tcpip_{DD4B2C91-0BA3-49E0-8AF5-  
E08A-000000000000} Media disconnected
```

```
Device\Tcpip_{26C4D5E5-CA1D-4554-B7D3-  
E08A-000000000000}
```

HUSHNANDAN KUMAR>ipconfig

Configuration

Adapter Ethernet 4:

tion-specific DNS Suffix . :
Local IPv6 Address : fe80::c629:1ac2:2069
Address : 192.168.56.1
Mask : 255.255.255.0
Gateway :

LAN adapter Local Area Connection* 1:

State : Media disconnected
tion-specific DNS Suffix . :

LAN adapter Local Area Connection* 2:

State : Media disconnected
tion-specific DNS Suffix . :

Adapter Bluetooth Network Connection:

State : Media disconnected
tion-specific DNS Suffix . :

LAN adapter Wi-Fi:

tion-specific DNS Suffix . :
Local IPv6 Address : fe80::6bf4:5171:ccfc
Address : 10.85.19.170
Mask : 255.255.248.0
Gateway : 10.85.16.1

Q6: Explain observations

Difference Between Wired and Wireless Speed:-

↳ Wired network is:

Faster

More stable

Less signal interference

↳ Wireless network is:

Portable

Easy to connect

Slightly slower due to signal loss.

Role of Subnet Mask

↳ Subnet mask identifies.

↳ Network portion

↳ Host portion of IP address

↳ Helps devices communicate within same network.

Eg. IP: 192.168.1.5

Subnet Mask: 255.255.255.0